SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXP:10

Name: Balaguru S

Register Number: 192312070

Test Plan Document for a Library Management System

Aim

To develop a comprehensive Test Plan for a Library Management System, covering the test scope, test strategy, test design techniques, test execution strategy, environment, schedule and test completion criteria.

Problem Statement

A library management system keeps records of books and members, issues and returns books, calculates fines and sends reminders. Wrong stock counts, wrong due dates or wrong fines upset members and make the library records unreliable.

Testing is needed to check that the application:

- Books and members are recorded and searched correctly.
- Issue, return, renewal, reservation and fine rules work as specified.
- Roles and access rights are enforced.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The librarian and members log in with role-based access.
FR2	The librarian can add, update and delete book records (title, author, ISBN, copies). The ISBN must be unique.
FR3	The librarian can register members with name, member ID, department and contact details.
FR4	Users can search books by title, author or ISBN.
FR5	A member can borrow up to 3 books at a time for 14 days.
FR6	When a book is returned, the system updates the number of available copies.
FR7	A fine of ₹2 per day is charged after the due date.
FR8	A member can renew a book once if no one else has reserved it.
FR9	A member can reserve a book that is currently issued.
FR10	A member with unpaid fines, or who already holds 3 books, cannot borrow more books.
FR11	Email reminders are sent 2 days before the due date and on the due date.
FR12	The librarian can generate reports of issued, overdue and most borrowed books.

2 Software and environment requirements

Item	Details
Application	Web application (Java/PHP back end, HTML/JavaScript front end)
Database	MySQL (books, members, transactions, fines)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	50 book records, 20 member records, issued and overdue transactions, sample email account

Test Plan

1 Test plan details

Item	Details
Project	Library Management System
Document	Test Plan
Version	1.0
Prepared by	Balaguru S (192312070)
Date	01-11-2026
Reference documents	Software Requirements Specification (functional requirements FR1 to FR12 above)

2 Test objectives

- Verify that every functional requirement is implemented correctly and can be traced to at least one test case.
- Find defects early and report them with enough detail to fix them.
- Check non-functional qualities: usability, performance, security and compatibility.
- Provide clear completion criteria so that the release decision is based on test results.

3 Test scope

In scope:

- Login and role-based access for librarian and member.
- Book and member management (add, update, delete, search).
- Issue, return, renewal and reservation of books.
- Fine calculation and borrowing restrictions.
- Email reminders and reports.

- Usability, compatibility and basic security checks.

Out of scope:

- Hardware such as barcode scanners and RFID gates.
- Third-party email server performance.
- Payment gateway for online fine payment (if added later).
- Data migration from the old paper records.

4 Test strategy and test levels

Test level	Focus	Performed by	Tools / method
Unit testing	Each function or class works on its own	Developers	JUnit / NUnit, code review
Integration testing	Modules work together (UI, business logic, database)	Developers and testers	Postman, API tests
System testing	Complete application against all requirements	Test team	Manual tests, Selenium
Regression testing	Old features still work after changes or fixes	Test team	Automated Selenium suite
Non-functional testing	Performance, security, usability, compatibility	Test team	JMeter, browser checks
User acceptance testing	Users confirm the system meets their needs	End users and client	Business scenarios

5 Test design techniques

Technique	Applied to	Example
Equivalence partitioning	Search input, member type, ISBN	Valid ISBN, invalid ISBN, blank ISBN
Boundary value analysis	Borrow limit, loan period, fine	Borrow 2, 3 and 4 books; return on day 14 and day 15
Decision table	Borrowing rules	Combinations of fine pending, books held and copies available
State transition	Book status	Available → Issued → Reserved → Returned
Error guessing	Forms and search	Duplicate ISBN, special characters, very long titles

Requirement-to-test coverage:

Req. ID	Test condition	Technique
FR1	Login with valid and invalid credentials; access to librarian pages by a member	Equivalence partitioning
FR2	Add a book with a duplicate ISBN; update and delete a book	Error guessing
FR3	Register a member with valid and missing details	Equivalence partitioning
FR4	Search by title, author and ISBN; search with no match	Equivalence partitioning
FR5	Borrow the 3rd and the 4th book; check the due date	Boundary value analysis
FR6	Return a book and check the available copies	State transition
FR7	Return a book on day 14, 15 and 20 and check the fine	Boundary value analysis
FR8	Renew a book once, then twice; renew a reserved book	Decision table
FR9	Reserve an issued book; reserve an available book	State transition
FR10	Borrow with a pending fine; borrow with 3 books held	Decision table
FR11	Check reminder emails 2 days before and on the due date	Equivalence partitioning
FR12	Generate each report and compare with the database	Equivalence partitioning

6 Test execution strategy

Testing is carried out in cycles. Each cycle starts when the build meets the entry criteria and ends when the cycle goals are reached.

Cycle	Tests executed	Goal
Cycle 1: Smoke	Login and the most important flow of the application	Confirm the build is stable enough for detailed testing
Cycle 2: Functional	All positive and negative test cases for FR1 to FR12	Find functional defects and report them
Cycle 3: Regression	Re-test fixed defects and run the regression suite	Confirm fixes and make sure nothing else is broken
Cycle 4: Non-functional	Performance, security, usability and compatibility tests	Check quality attributes against the targets
Cycle 5: UAT	Business scenarios run by end users	Get user sign-off for release

Execution rules:

- Test cases are executed in priority order: high, medium, low.

- Each result (Pass, Fail, Blocked) is recorded, and every failure is logged as a defect with steps, expected and actual result.
- Fixed defects are re-tested, and the related areas are regression tested.
- A daily status report shows the test cases run, passed, failed and blocked, and the open defects.

7 Test environment

Item	Details
Application	Web application (Java/PHP back end, HTML/JavaScript front end)
Database	MySQL (books, members, transactions, fines)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	50 book records, 20 member records, issued and overdue transactions, sample email account

8 Roles and responsibilities

Role	Responsibility
Test manager	Prepares the test plan, tracks progress, reports to stakeholders
Test engineer	Writes and runs test cases, logs defects, re-tests fixes
Automation engineer	Builds and maintains the automated regression suite
Developer	Fixes defects and supports unit and integration testing
Business analyst / client	Clarifies requirements and gives UAT sign-off

9 Test schedule

Activity	Duration	Responsible
Test planning and requirement review	1 week	Test manager
Test scenario and test case design	2 weeks	Test engineers
Environment and test data setup	1 week	Test team, developers
Cycles 1 and 2: smoke and functional testing	3 weeks	Test engineers
Cycle 3: regression testing	1 week	Test and automation engineers
Cycle 4: non-functional testing	1 week	Test team
Cycle 5: user acceptance testing	1 week	End users, test manager

Activity	Duration	Responsible
Test summary report and sign-off	2 days	Test manager

10 Test completion criteria

Entry criteria:

- The requirements are approved and the test cases are reviewed.
- The build is deployed in the test environment and passes the smoke test.
- Test data and test accounts are ready.

Exit (completion) criteria:

- 100% of the planned test cases are executed.
- At least 95% of the executed test cases pass.
- No Critical or High severity defect is open.
- All functional requirements are covered by at least one executed test case.
- The regression suite passes on the final build, and the UAT sign-off is received.

Suspension and resumption:

- Testing is suspended if the build is not stable (for example, login fails) or the test environment is down.
- Testing resumes when a new build passes the smoke test or the environment is restored.

11 Risks and mitigation

Risk	Likelihood	Impact	Mitigation
Fine calculation errors because of date and time-zone handling	Medium	High	Test with boundary dates; compare with manual calculations
Late changes to borrowing rules	Medium	Medium	Keep the test cases linked to requirement IDs and update them quickly
Email server not available for reminder tests	Medium	Low	Use a mail test server (mock SMTP)
Limited test data for overdue cases	Low	Medium	Prepare data scripts to create overdue records

12 Defect management

Severity	Meaning	Target action
Critical	System crash, data loss or security breach; no workaround	Fix before the next build
High	Major feature fails; no easy workaround	Fix in the current cycle

Severity	Meaning	Target action
Medium	Feature partly fails; a workaround exists	Fix before release
Low	Cosmetic or minor issue	Fix if time permits

Defect life cycle: New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still present after retesting, it is reopened and sent back to the developer.

13 Test deliverables

- Test plan (this document)
- Test scenarios and test cases, with the requirement traceability matrix
- Test data and test environment details
- Defect reports and daily status reports
- Test summary report

Result / Conclusion

Result: A complete test plan was prepared for the Library Management System. It covers the test scope, 6 test levels, 5 test design techniques, a requirement-to-test coverage table for all 12 functional requirements, a 5-cycle execution strategy, the environment, roles, schedule, completion criteria, risks and defect management.

Conclusion: The test plan gives the Library Management System project a clear and repeatable way of testing. Testing against fixed completion criteria, with every requirement traced to tests, helps find defects early and supports a reliable release decision. The plan should be reviewed and updated whenever the requirements or the schedule change.